

Dinesh Kannan

+1 919-633-5438 | dineshkannan010@gmail.com | linkedin.com/in/dinesh012 | Dineshkannan010.github.io

EDUCATION

North Carolina State University

August 2023 - May 2025

Masters of Computer Science

GPA: 3.85/4.0

Coursework: Automated Software Engineering, Cloud Computing, Neural Networks, Object Oriented Design and Development, Database Management Systems, Design and Analysis of Algorithms

WORK EXPERIENCE

North Carolina State University | Researcher - Software Development

August 2025 - Present

- Built full-stack geospatial platform with **React, Node.js, PostgreSQL** serving 100+ users, implementing a custom Timeline Builder with before/after image comparison sliders that increased student research engagement by **35%**.
- Implemented **RBAC authentication** system with **JWT** and role-based workflows, reducing instructor approval time by **40%**.
- Reduced **page load times by 75%** (3.2s $\rightarrow$ 0.8s) as measured across multiple simultaneous user sessions, by migrating media infrastructure to **Cloudinary** CDN and offloading 60% of database storage operations.
- Designed automated draft-to-publish pipeline handling timeline submissions in a semester with **99% accuracy**.

North Carolina State University | Research Assistant

July 2024 - April 2025

- Developed a responsive full-stack application (**React, Flask, MongoDB**) with **JWT** authentication, query caching, and vector similarity search; integrated **LangChain** for orchestration and **deployed on AWS** with CI/CD automation.
- Architected end-to-end **RAG-powered AI Teaching Assistant** using OpenAI GPT-4 and vector embeddings, serving 500+ students with **35%** improvement in resource accessibility through semantic search and context-aware responses.
- Designed and deployed agentic workflow** with human-in-the-loop feedback loops, applying prompt engineering techniques to optimize LLM accuracy by **30%** and reduce hallucination rates through iterative context refinement.
- Implemented **LLM observability and monitoring** using custom logging pipeline to track response quality, latency (avg 1.2s).
- Sample Efficient Rapid Thermal Modeling of 3D IC:** Developed a **conditional diffusion model** in **PyTorch** for thermal map prediction in 3D ICs, lowering error by 61% and hotspot deviation by 2.47 pixels compared to baseline models.
- Optimized model architecture, lowering temperature error to 1.21°C with 500 samples, which improved modeling of complex chip layouts using limited data, leading to **publishing as a research paper in IEEE EPEPS 2025**.

Deloitte | Deloitte Consulting Analyst

January 2023 - July 2023

- Developed **.NET plugins** for **Dynamics 365**, reducing query response time by **40%** (8s $\rightarrow$ 4.8s) for 10,000+ record datasets.
- Identified critical vegetation growth patterns by **analyzing 50GB+ of datasets** using Python & SQL, directly influencing client resource allocation strategy and reducing projected maintenance costs by 15%.
- Built custom business automation workflows** with Power Automate and Azure Data Factory, streamlining cross-system data processing for 5 client projects and reducing manual effort by 50 hours/month.
- Accelerated **feature delivery by 20%** by leading daily stand-ups and refining backlog grooming, successfully **shipping 10 core features across 6 sprints** with zero post-deployment defects.

TECHNICAL SKILLS

Programming & Frameworks: Python, JavaScript, ReactJS, Node.js, Flask, .NET, TypeScript, Rest APIs, GraphQL

Cloud & Databases: AWS Cloud, Docker, NoSQL, MySQL, PostgreSQL, Google Cloud, Atlas Vector Search

Tools: Git, Jira, Agile/Scrum, Postman, MLflow, DVC, TDD, Linux, CI/CD (GitHub Actions, Docker), Azure Data Factory

AI & ML : LLMs, Agentic AI, PyTorch, Tensorflow, RAG, OpenAI, LangChain, LangGraph, CNNs, Transformers

PROJECTS

Real-Time YouTube Sentiment Analysis via Chrome Extension | MLflow, DVC, AWS, Flask

- Developed an **end-to-end MLOps pipeline** from data collection to cloud deployment, integrating MLflow, DVC, and Docker to analyze YouTube sentiment in real-time through a custom-built Chrome extension.
- Deployed scalable inference API** on AWS Lambda with Docker containerization, handling 500 requests/hour with 200ms average latency; integrated DVC for dataset versioning and MLflow for experiment tracking.

Ellipsis: Podcast Generation Platform | Python, Flask, React, Redis, Orpheus TTS

- Automated podcast generation achieving 95% natural-sounding speech quality as measured by user feedback surveys, by integrating **Perplexity's deep research API with Orpheus TTS** to produce multi-topic content in under 3 minutes per episode.
- Engineered a modular **multi-agent system** using **MCP architecture** to orchestrate 5 specialized agents for research, debate, and TTS workflows.